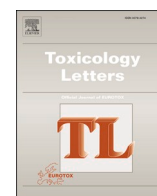




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Cadmium physiologically based pharmacokinetic (PBPK) models for forward and reverse dosimetry: Review, evaluation, and adaptation to the U.S. population

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ABSTRACT

The goal of this study was to assess a cadmium (Cd) physiologically based pharmacokinetic (PBPK) model to evaluate Cd toxicological reference values (e.g. reference dose, tolerable intake, minimum risk level) adapted to the U.S. population. We reviewed and evaluated previously published Cd PBPK models and developed further adaptations to the 1978 Kjellström and Nordberg (KN) model. Specifically, we propose adaptations with updated U.S.-specific bodyweight, kidney weight and creatinine excretion models by using NHANES data as well as a stochastic PBPK model that provides credible intervals of uncertainty around mean populational estimates. We provide our model review and adaptations as well as present estimates from the newly adapted models using observed U.S. urinary Cd values as a function of gender and age and given dietary exposure as evaluated from NHANES/WWEIA and U.S. Total Diet Study data. Results show all newly adapted models provide acceptable mean estimates of urinary Cd in the U.S. The stochastic model provides credible intervals to further inform regulatory decision making. Validation of the estimated K-Cd concentration values was not possible as data for a representative population was not available. We developed a web-based tool implementing these models and other potential adaptations to facilitate PBPK model estimate comparisons.

1. Introduction

Cadmium (Cd) is an environmental contaminant that has long been recognized to pose risks to human health with dietary exposure being the most likely oral route (ATSDR, 2012; Schaefer et al., 2020). The sensitive target organs of oral exposure to Cd have been identified as the kidney and the bone (ANSES, 2017; ATSDR, 2012; EFSA, 2009; JECFA, 2011). Åkesson et al. (2005) reported urinary Cd (U-Cd) levels of 0.67 µg/g creat to be significantly associated with effects on renal tubules and Engström et al. (2011) found a decrease in bone mineral density at a median U-Cd of 0.87 µg/g creat. Evidence in laboratory animals also suggests that immunological, neurological, reproductive, and developmental toxicity may occur after oral Cd exposure (ATSDR, 2012) but the initial confidence in the body of literature is low (Schaefer et al., 2022). The United States Food and Drug Administration's (FDA) Closer to Zero (C2Z) action plan (U.S. Food and Drug Administration, 2021a) puts forward a science-based, iterative approach to reduce dietary exposure

to toxic elements, through regulatory actions, stakeholder engagement and research. The plan includes evaluating existing data and scientific information, and through a process of engagement and consultation, establishing interim reference levels (IRL; a maximum daily intake from food) or toxicological reference values (TRVs), that can be used to guide action levels for toxic elements in food, when appropriate. The research presented here in reviewing, evaluating, and ultimately developing a PBPK model contributes to that effort. The PBPK model could also be useful in evaluating the potential impact of changes in exposure from food arising from changes in contamination or consumption.

Several PBPK models have been published for Cd (Amzal et al., 2009; Kjellström and Nordberg, 1978; Maza et al., 2021; Sheng et al., 2000) of which the Kjellström and Nordberg (1978) model and the Amzal et al. (2009) model have been used by national or international agencies (ANSES, 2017; ATSDR, 2012; EFSA, 2009; JECFA, 2011). None of the original models characterize uncertainty, though the Amzal et al. (2009) model was intended to include individual variability. The goal of this

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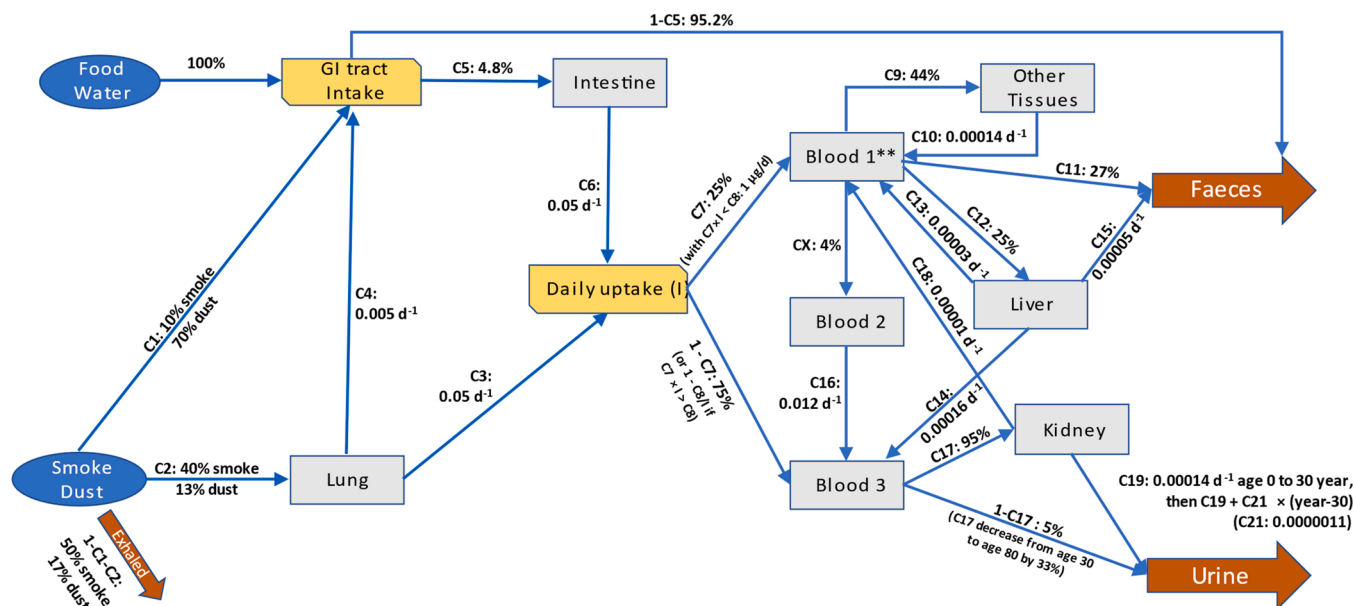


Fig. 1. The original KN model (Kjellstrom and Nordberg, 1978). Blood 1 is the plasma compartment where Cd may bind to plasma components (i.e., albumin and other organic constituents). Blood 2 is the red-blood cell compartment, which represents the accumulation of Cd in erythrocytes. Blood 3 represents the binding of Cd to metallothionein. For mass balance purposes, an arrow from GI tract Intake to Faeces was added as well as the “exhaled” excretion route.

study is to develop a Cd PBPK model specific for the U.S. population. To achieve this goal, we reviewed and evaluated available models, adapted model components to capture the characteristics of the U.S. population, integrated parameter uncertainty estimates to provide credible intervals around mean estimates, and implemented an algorithm to properly estimate exposure and credible intervals when employing reverse dosimetry with the stochastic model. We also created a web-based tool to compare estimates from models with different parametrizations and assumptions and for different subpopulations.

2. Material and methods

2.1. PBPK models used by national and international organizations

2.1.1. The Kjellstrom and Nordberg (1978) model

Kjellstrom and Nordberg (1978) developed a PBPK model for Cd hereinafter denoted as the “KN model”. It is an 8-compartment model with 21 parameters fitted manually to Cd concentrations as observed in blood, kidney cortex, liver and urine of Swedish and Japanese populations (see Fig. 1 and the corresponding set of differential equations in the supplemental material [Table S1]; Table S2 additionally includes model parameter details). The KN model does not consider individual variability or parameter uncertainty. In their original report, the authors indicated they included assumed ranges on the various coefficients but cautioned users that these ranges should not be considered as variability ranges or uncertainty ranges without further inferences from data. As such, the values obtained from the KN model indicate an approximate mean value that would be expected in a group of people exposed to a given Cd level. The KN model is a deterministic population model where a single input leads to a single output.

2.1.2. The Amzal et al. (2009) model

To estimate population variability using the KN model, the European Food Safety Agency (EFSA, 2009) developed and used the Amzal et al. (2009) model, a simplified version of the KN Model (from 8 to 1 compartment) that links dietary Cd intake (in $\mu\text{g}/\text{kg}$ body weight (BW)/day) to U-Cd (in $\mu\text{g}/\text{g}$ creatinine). The model provides population variability distribution of the half-life parameter estimated through a Bayesian hierarchical model.

We reproduced the Amzal et al. (2009) model and made inferences on the model parameters using a Bayesian approach and U.S. data on Cd dietary intake and U-Cd from the National Health and Nutrition Examination Survey (NHANES). However, the Amzal et al. (2009) model showed inconsistencies where the predicted mean estimates aligned well with the observed data but the 90th percentile estimates were vastly different from the observed values. Upon further investigation, we traced the discrepancies to a correlation between Cd intake and the estimated half-life values in the inference process. We were unsuccessful in obtaining the original model and/or the data used to derive it, so it was not possible for us to determine whether the discrepancy identified resulted from the use of U.S. data or is a general issue with the model. As a result, we do not consider this model or its adaptations further.

2.1.3. Other published adaptations of the Kjellstrom and Nordberg (1978) model

The KN model has been used and slightly modified by various other groups (Béchaux et al., 2014; Choudhury et al., 2001; Diamond et al., 2003; Fransson et al., 2014; Leconte et al., 2021; Qing et al., 2021; Ruiz et al., 2010).

Choudhury et al. (2001) adapted the KN model in terms of growth algorithms for females and males (using the O’Flaherty, 1993 formula as a function of age and gender), Cd concentration in the kidney cortex (calculated based on Cd burden in kidney and kidney weight), and rate of creatinine excretion (based on its relationship with lean body mass using a weighted arithmetic mean from eight studies reported in Forbes and Bruining, 1976). Additionally, while the original KN model C5 parameter (fraction of GI-tract Cd retained in epithelium) was estimated as 4.8% for both women and men (Kjellstrom and Nordberg, 1978), Choudhury et al. (2001) suggested a gender difference and proposed 5% for men and 10% for women given the shown increased intestinal absorption for women caused by lower body iron stores (Järup and Åkesson, 2009).

These newly proposed C5 values by Choudhury et al. (2001) were implemented by Diamond et al. (2003) and Ruiz et al. (2010) using the rest of the model parameters as in Choudhury et al. (2001). A sensitivity analysis conducted on the KN model input parameters (excluding the inhalation pathway) indicated that the rates for oral absorption (C5) was, by far, the input parameter that most affected the model output

(Dede et al., 2018; Ruiz and Fowler, 2015).

Béchaux et al. (2014) included lower and upper ranges of the values provided by Kjellstrom and Nordberg (1978) and Ruiz et al. (2010) as individual variability in triangular distributions. Based on inferences on the French population showing less impact of gender on absorption, Béchaux et al. (2014) proposed and used C5 values of 5% for males (3%–10% range) and 7.5% for females (5%–10% range).

Fransson et al. (2014) were able to make more robust inferences on the KN model parameters C7, C8, C16, C17, C19 and C20 as the authors had matched kidney cortex, blood and urinary Cd measurements obtained from 82 women and men (30 never smokers). They provided uncertainty bounds around the mean values for those parameters. By providing credible intervals of some of the most sensitive parameters of the KN model (Table S3), uncertainty measures around the output of the KN model can be derived via Monte Carlo simulation.

Qing et al. (2021) provided seven updated Cd kinetic parameters from the KN model through Bayesian computation solely based on one pair of Chinese dietary intake and U-Cd data for each individual in their survey.

Leconte et al. (2021) used the modified version of the original KN model as described in Ruiz et al. (2010) and adapted body weight and creatinine excretion levels to those based on French population data.

2.2. Our proposed adaptations to the KN model for use on the U.S. population

2.2.1. Body weight

Most of the previous uses of the KN model (ATSDR, 2012; Choudhury et al., 2001; Diamond et al., 2003; Ruiz et al., 2010) used the O'Flaherty (1993) model to estimate the bodyweight of the population as a function of age and gender. We propose using a specific growth model for the U.S. population similar to that done by Leconte et al. (2021) in France. We collected bodyweight data from the 2003–2018 NHANES survey cycles (76,160 observations), calculated the weighted mean of the recorded body weights for each gender and each age (0- to 79-year-old) (CDC, 2021) and derived an empirical body weight model by fitting a loess function on these estimates. NHANES does not report the exact age of individuals aged ≥ 80 years, hence our model is limited to individuals under 80.

2.2.2. Kidney weight

Kidney weight is needed to translate the Cd kidney burden (in μg) from the KN model to a concentration of Cd in the kidney cortex ($\mu\text{g Cd/g kidney cortex}$). O'Flaherty (1993) developed a kidney weight model of the form:

$$kw(\text{age}, \text{gender}) = kw_a(\text{gender}) \times \left(\frac{BW(\text{age}, \text{gender})}{BW_a(\text{gender})} \right)^{0.84}$$

where $kw(\text{age}, \text{gender})$ is the average weight of the kidneys for an individual of age age and gender gender , and $kw_a(\text{gender})$ and $BW_a(\text{gender})$ are the average weights of the kidneys and body weight of adults of corresponding gender, respectively.

We used $kw_a(\text{male}) = 310 \text{ g}$ and $kw_a(\text{female}) = 275 \text{ g}$ from ICRP (1981) and the adult bodyweight data estimated from NHANES (an average weight of 88.0 kg for 20–40 year old males and 75.6 kg for 20–40 year old females). Fridberg et al. (1974) and Choudhury et al. (2001) reported that Cd concentrations in the renal cortex (rcc , $\mu\text{g/g}$) are 1.5 times the average concentration in the kidney. The Cd concentration in kidney cortex (K-Cd) was then calculated as:

$$rcc = 1.5 \frac{K(\text{age}, \text{gender})}{kw(\text{age}, \text{gender})},$$

where $K(\text{age}, \text{gender})$ is the renal Cd burden (μg) estimated from the KN model.

Table 1
Cadmium PBPK models evaluated where * indicates models including adaptations proposed in this study.

Model # Model Type	KN Model* parameters and adaptations	Body Weight Model	Creatinine Excretion Model
#1 Deterministic	Original (Table S2) with C5: 5% for male and 10% for female as described in Choudhury et al. (2001)	O'Flaherty (1993) as described in Choudhury et al. (2001)	From Forbes and Bruining (1976) as described in Choudhury et al. (2001)
#2* Deterministic	Original (Table S2) with C5: 5% for male and 7.5% for female as in Béchaux et al. (2014)	Empirical model using NHANES data	Ix et al. (2011) "Equation D" applied to NHANES data
#3* Stochastic	Original (Table S2) with C5: Pert ¹ (3, 5, 10%) for males and Pert(3, 7.5, 10%) for female; C7: Pert2(28.8, 40.4, 52.7%), C8: set to 1000, C16: Pert2(0.0144, 0.0209, -0.0291) day ⁻¹ , C17: Pert2(82.0, 92.2, 99.1%), C19: Pert2(0.0000219, 0.0000473, 0.0000774) day ⁻¹ , and C20: Pert2(3.81, 5.30, 7.36%) and C21 = 0.0000011 / 0.00014 $\times$ C19 as estimated in Fransson et al. (2014) (Table S3)	Empirical model using NHANES data	Ix et al. (2011) "Equation D" applied to NHANES data

^a Kjellstrom and Nordberg (1978) model;

^b Pert distribution Pert(a, b, c) with min value a, most probable value b and maximum value c (Vose, 2000);

^c Pert distribution Pert(a, b, c) with min value a, median value b and maximum value c.

2.2.3. Creatinine excretion

Choudhury et al. (2001) developed a simple formula linking lean body mass and creatinine excretion using a weighted arithmetic mean from eight studies reported in Forbes and Bruining (1976). However, creatinine excretion as a function of lean body mass has been said to vary according to gender and age (Barr et al., 2005). We propose using a different model of creatinine excretion (g per day) that is adapted to the U.S. population with regards to age, gender, bodyweight, and race. We used an equation (“Equation D”) developed by the Chronic Kidney Disease (CKD) Epidemiology collaboration (Ix et al., 2011) which was validated for men and woman over 20 years of age in the U.S. and used the BW and age data as collected from NHANES:

$$Creat = 879.89 + 12.51 \times BW - 6.19 \times age + (34.51 \text{ if Black}) - (379.42 \text{ if female})$$

where *Creat* is the creatinine excretion in g creatinine per day, body weight (*BW*) is in kg and *age* is in years.

2.2.4. Model evaluation

We evaluated three potential adaptations of the KN model most relevant to the U.S. population (Table 1) for estimating U-Cd and K-Cd concentrations as a function of age and gender. These three models can be regarded as different parametrizations of the KN model. Model #1 is the original KN model as parameterized for the U.S. by Choudhury et al. (2001) where *C5* is set to 5% for males and 10% for females, the body weight model is from O’Flaherty (1993) and the creatinine excretion model is from Forbes and Bruining (1976). This model is the one used by Diamond et al. (2003) and Ruiz et al. (2010) described in Choudhury et al. (2001). Model #2 is the original KN model as parameterized by Béchaux et al. (2014), where *C5* is set to 5% for males and 7.5% for females, and the body weight, kidney weight, and creatinine excretion models are our newly proposed adaptations. Model #3 is similar to Model #2 but also includes our newly proposed Cd transfer from kidney to urine adaptation and implements uncertainty distributions for parameters *C7*, *C8*, *C16*, *C17*, *C19*, and *C20*, specifically Pert distributions (Vose, 2000) with minimum the lower bound of 2.5th quantile, median the 50th quantile and maximum the 97.5th quantile of the posterior distributions of each parameter as provided by Fransson et al. (2014) and an uncertainty distribution for parameter *C5* from Béchaux et al. (2014). We did not use the Fransson et al. (2014) *C5* derivation because their method underestimated the absorbed fraction of Cd (results not shown). In the original KN model, the rate of Cd transfer from kidney to urine (*C19* parameter, 0.00014 Day^{-1}) was assumed to increase with age for individuals > 30 years old by $C21 = 0.0000011 \text{ Day}^{-1}$ per year. We assumed in model #3 that the *C21* parameter was related to the *C19* parameters drawn from the Fransson et al. (2014) uncertainty distribution in the same proportion as *C21* and *C19* are related in the original KN model, that is, for each value of *C19*_{*i*} drawn from the Pert distribution, we used $C21_i = C19_i \times 0.0000011 / 0.00014$. For stochastic Model#3, we developed an algorithm to properly estimate exposure and credible interval for a given K-Ud / U-Cd level by reverse dosimetry using Monte-Carlo simulation.

2.3. Model validation

The evaluated models were validated on U.S. population data by comparing observed U-Cd with the values predicted by each model. Validation of the estimated K-Cd concentration values was not possible as data is not available.

2.3.1. Dietary exposure

The Cd dietary exposure from individuals surveyed during NHANES 2003–2004 and 2013–2014 cycles was estimated using consumption data from What We Eat in America (WWEIA), the food consumption survey portion of NHANES, and Cd contamination data from the Total

Diet Study (TDS) 2003–2004 and 2014–2016 (U.S. Food and Drug Administration, 2021b). The method was similar to the one used for children’s exposure to lead and Cd using the TDS 2014–2016 data (Spungen, 2019), in the hybrid scenario where Cd concentrations for samples below the limit of detection (LOD) were set to 0 for foods for which no sample above the LOD was detected and Cd concentrations for samples below the LOD were set to half the LOD if at least one sample over the LOD was observed. The two-day average dietary Cd exposure (μg/day) for each of the 15,732 individuals was estimated and divided by the individual’s body weight (kg).

2.3.2. Urinary cadmium

Observed U-Cd concentration data were collected from NHANES 2003–2004–2013–2014 cycles. U-Cd was obtained from spot urine samples using inductively coupled plasma-mass spectrometry (ICP-MS) and standardized by dividing the reported values by the creatinine concentration obtained from the same sample. Due to analytical method changes in 2007, creatinine levels from surveys prior to 2007 were adjusted according to CDC’s recommendations (CDC, 2021). We excluded ever smokers (declared having smoked at least 100 cigarettes during a lifetime) and individuals potentially exposed to Cd in the workplace (current job or the longest job in “mining”, “construction” or “manufacturing”). We considered children under 12 as non-smokers and individuals under 19 as not exposed to Cd from the workplace (8987 individuals remaining).

NHANES and TDS data analyses were done using the R (v 4.1.0, R Development Core Team, 2021) survey package (Lumley, 2004). A complete description of NHANES survey design and data collection is available in CDC (2021). PBPK models were coded in R and C++ and integrated using the *odeintr* package (Keitt, 2017). The main code and the algorithm for the reverse dosimetry derivation through Monte-Carlo simulation in model #3 is provided in the supplementary material.

2.4. Model implementation: cadmium forward and reverse dosimetry tool

We implemented the models in an open access web-based tool (R shiny app, Chang et al., 2021) to readily illustrate the output of the models, quickly compare estimates and conduct forward and reverse dosimetry analyses. The tool also allows the user to freely change PBPK parameter assumptions. In this paper, we provide an example of using the tool for forward and reverse dosimetry analyses and the results when using the different models evaluated here given a set of chosen input values. In the tool, the user can assume a constant exposure to Cd from birth until the age under evaluation or can provide the data as intake per kg BW for each year of age. The link to access the tool is: https://fda-riskmodels.foodrisk.org/Cadmium_FDA_PBPK/.

3. Results and discussion

3.1. Model evaluation

We collected data on dietary exposure as well as urinary cadmium from NHANES (CDC, 2021).

Mean Cd dietary intake in μg/kg BW/day in the U.S. (data from 15,732 individuals) decreased with increasing age, where females 6–11-, 20–39-, and 60–79-year-old had means of 0.267 (CI 95%: 0.254; 0.281), 0.142 (CI 95%: 0.137; 0.147) and 0.133 (CI 95%: 0.127; 0.140) μg/kg BW/day, respectively. Similar means and trends were found for male subjects with mean intake values being slightly higher in all age categories (Table S4 and Table S6, Fig. S1). While mean Cd intake per kg bodyweight sharply decreased with increasing age from age 0–20, it was relatively constant for the different survey years and age and gender for individuals over 20 years old (Fig. S1). This decreasing trend in dietary intake per bodyweight with age for individuals under 20 arises primarily from the rapid change in bodyweight with age while Cd intake (in g) did not differ substantially. The half-life of Cd is 10–30 years in humans

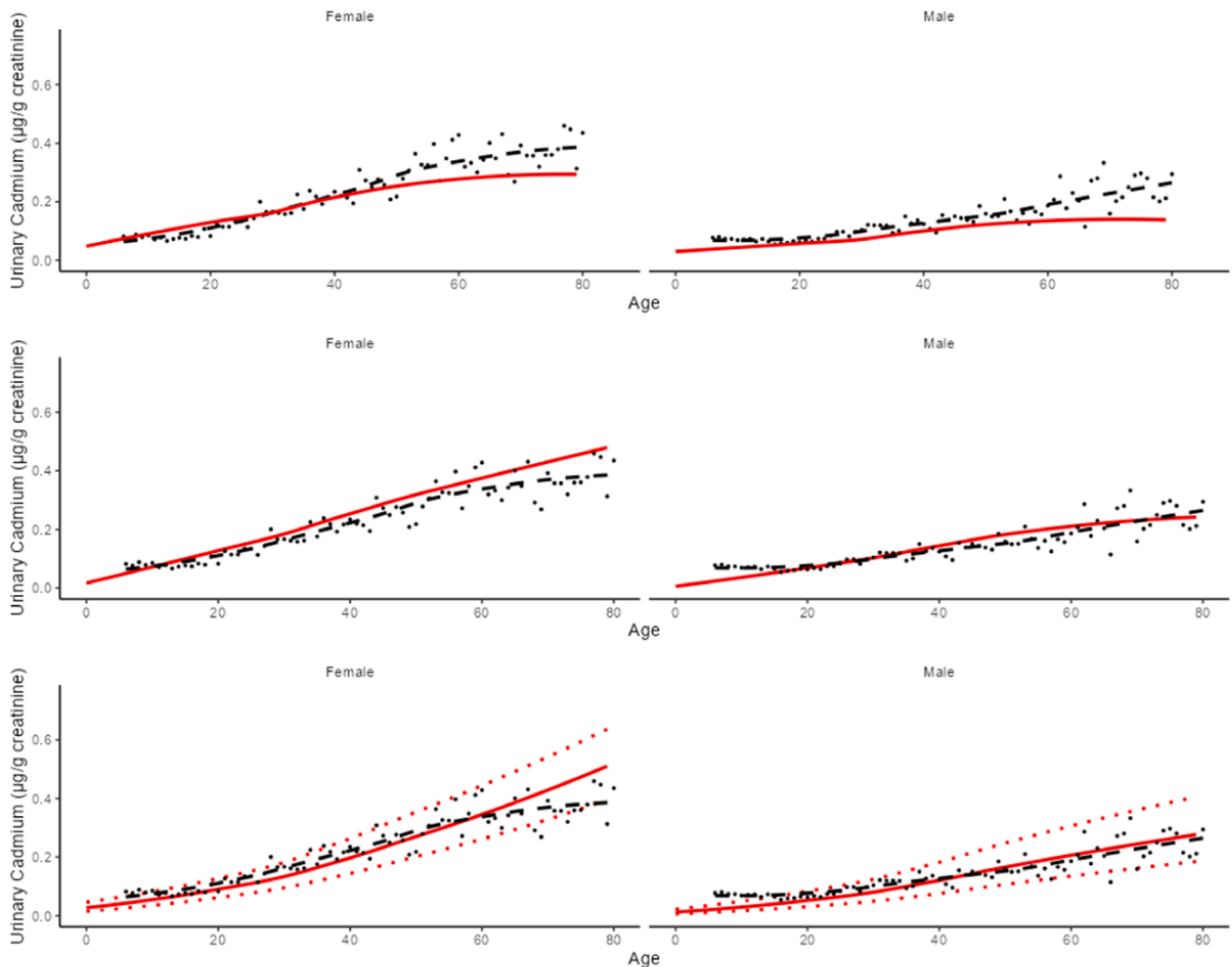


Fig. 2. Urinary Cd ($\mu\text{g/g creatinine}$) by age and gender. NHANES survey 2003–2004–2013–2014 for never smoker and not occupationally exposed. (Black dots: observed geometric mean for each year, dotted black line: smooth regression). Red line: output from the Kjellstrom and Nordberg (1978) model under the assumption of a mean dietary exposure each year of age equal to the geometric mean estimated from NHANES survey 2003–2004 and 2013–2014 and TDS data. Top: original KN model as described by Choudhury et al. (2001) (Model #1); middle: KN model with C5 parameter as modified by Béchaux et al. (2014), specific growth model and creatinine excretion model (Model #2); bottom: KN model with C5 parameter as modified by Béchaux et al. (2014), specific growth model and creatinine excretion model, C7, C8, C16, C17, C19 and C20 parameters uncertainty as estimated by Fransson et al. (2014) and C21 parameter proportional to C19 parameter (pointed line: 5th and 95th quantile of uncertainty) (Model #3). See text for further details.

(EFSA, 2009) so much of the Cd intake (i.e. >87.5% for a half-life of 10 years and >50% for half-life of 30 years) during the first 20 years of life is eliminated by the time individuals are in their 50 s

Observed U-Cd levels (in $\mu\text{g/g creatinine}$) in the U.S. estimated from NHANES (8987 individuals) significantly increased with age, where females 6–11-, 20–39, and 60–79-year-old had means of 0.080 (CI 95%: 0.076; 0.084), 0.154 (CI 95%: 0.146; 0.163) and 0.348 (CI 95%: 0.326; 0.372), respectively. On average, observed U-Cd levels were significantly higher in woman than men (Table S5 and Figure S2).

All three models evaluated (Table 1) provide estimates of the mean U-Cd from the dietary intakes in the population for men and women as a function of age (Fig. 2) that are close to the mean observed values. Model #1 underestimates the observed U-Cd concentrations, particularly in older adults whereas Model #2 tends to overestimate this concentration in older women. Without uncertainty estimates, it is not possible to determine whether these differences in predicted U-Cd levels are within the range of the observed values. The KN model with the Fransson et al. (2014) parametrization (model #3) provides credible intervals around the mean estimates and these generally include the observed data.

While models #1, #2 and #3 provide comparable estimates for U-Cd, model #3 using Fransson et al. (2014) parameter adaptations provides significantly higher estimates for Cd burden in kidneys as compared to model #1 or model #2 (Fig. 3). Fransson et al. (2014) pointed out that because their inferences arose from paired K-Cd concentrations and U-Cd from the same individuals, their estimates of the C19 parameter, linking observations in kidney with urinary excretion, were likely more accurate than the parameter estimations of Kjellstrom and Nordberg (1978) where the data used originated from unrelated studies and measurements. Akerstrom et al. (2013) evaluated the relationship between Cd in urine and kidney using the same dataset as Fransson et al. (2014) and found a 1:60 ratio for U-Cd:K-Cd. Model #1 underestimates the 1:60 U-Cd:K-Cd ratio to a similar degree that model #3 overestimates it, with model #2 greatly underestimating the U-Cd:K-Cd ratio (Table 2). Fransson et al. (2014) specifically noted that their new parameters lead to predictions of a higher cadmium burden in the kidney that were more accurate than the original KN model.

While all three models provide acceptable estimates of the mean U-Cd in the population, the bodyweight and creatinine excretion model

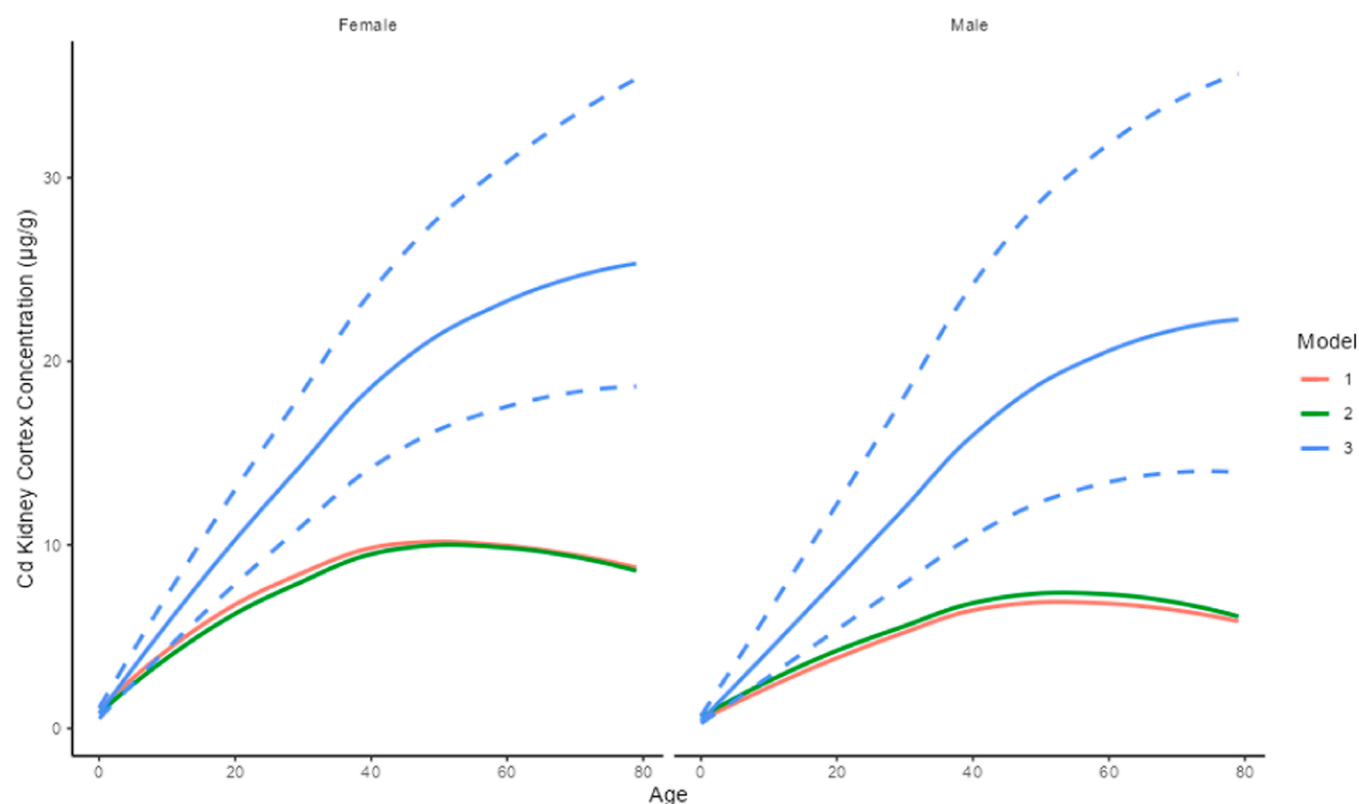


Fig. 3. Output from the Kjellstrom and Nordberg (1978) model for K-Cd ($\mu\text{g/g}$) as a function of age and gender under the assumption of a mean dietary exposure each year of age equal to the geometric mean estimated from NHANES survey 2003–2004 and 2013–2014 and TDS data. Red line: original KN model as modified by Choudhury et al. (2001) (Model #1); Green line: KN model with C5 parameter as modified by Béchaux et al. (2014), specific growth model and creatinine excretion model (Model #2); Blue lines: KN model with C5 parameter as modified by Béchaux et al. (2014), specific growth model and creatinine excretion model, and C7, C8, C16, C17, C19 and C20 parameters uncertainty as estimated by Fransson et al. (2014) and C21 parameter proportional to C19 parameter (dotted line: 5th and 95th quantile of uncertainty) (Model #3). See text for further details.

Table 2

Estimated mean urinary Cadmium (U-Cd) and Cadmium in the kidney cortex (K-Cd) with Model #1, #2 or #3 in 50-year-old women and men assuming a dietary intake of $0.2 \mu\text{g/kg BW/day}$ from birth up to 50 years of age (forward dosimetry, see text for details).

Measure	Women			Men		
	Model #1	Model #2	Model #3	Model #1	Model #2	Model #3
U-Cd	0.51 $\mu\text{g/day}$ (0.34 $\mu\text{g/g creat}$)	0.51 $\mu\text{g/day}$ (0.43 $\mu\text{g/g creatinine}$)	0.42 $\mu\text{g/day}$ [CI90%: (0.31; 0.56)] 0.36 [CI 90%: 0.26; 0.47] $\mu\text{g/g creat}$	0.33 $\mu\text{g/day}$ (0.15 $\mu\text{g/g creat}$)	0.39 $\mu\text{g/day}$ (0.23 $\mu\text{g/g creatinine}$)	0.35 $\mu\text{g/day}$ [CI90%: (0.23; 0.54)] 0.2 [CI 90%: 0.13; 0.32] $\mu\text{g/g creat}$
K-Cd	14.2 $\mu\text{g/g}$	13.5 $\mu\text{g/g}$	27.9 $\mu\text{g/g}$ [CI 90%: 21; 36] $\mu\text{g/g}$	7.9 $\mu\text{g/g}$	9.2 $\mu\text{g/g}$	20.5 $\mu\text{g/g}$ [CI 90%: 13.6; 30.3] $\mu\text{g/g}$
Note: estimated Creatinine excretion	1.51 g creat/day	1.18 g creat/day	1.18 g creat/day	2.12 g creat/day	1.71 g creat/day	1.71 g creat/day

estimates used differ significantly. Fig. 4 illustrates the O'Flaherty (1993) body weight model used by Choudhury et al. (2001) (and used in Model #1) compared to the observed mean body weight estimated from NHANES in the U.S. population (Table S6). The body weights in NHANES are significantly higher than those predicted by the O'Flaherty (1993) model. Additionally, NHANES data indicated a decrease in body weight for individuals over 60 years old while the modeled data with the O'Flaherty (1993) formula is approximately constant. Using the O'Flaherty (1993) model instead of our updated model would underestimate bodyweight which would lead to an underestimation of the Cd intake as expressed in $\mu\text{g/kg BW/day}$. This is part of the reason for the underestimation of U-Cd estimates observed with Model #1 in Fig. 2.

Fig. 4 also compares the updated U.S. creatinine excretion values (used in Models #2 and #3) to those by Choudhury et al. (2001) (Model #1) (Table S6). Using the proposed Ix et al. (2011) "Equation D" model

with NHANES bodyweight, age, gender and race information we model a decrease in creatinine excretion starting at age 30 which becomes sharper at age 60 while the Choudhury et al. (2001) models creatinine excretion to be approximately constant for individuals over 20 years old. The Forbes and Bruining (1976)'s data modeled by Choudhury et al. (2001) was validated only on a small populations while the Ix et al. (2011) model was validated for the U.S. population.

An output of the KN model is the quantity of Cd excreted in urine per day ($\mu\text{g/day}$). The concentration of Cd in urine measured in spot samples (e.g. NHANES data) is standardized to the excretion of creatinine in urine and expressed in $\mu\text{g/g creatinine}$. This standardization is needed because concentration of Cd in urine varies with urine dilution while the rate of creatinine excretion in urine is relatively constant for a given individual at a given time in life (Barr et al., 2005). Additionally, creatinine excretion and U-Cd are naturally indirectly linked because of

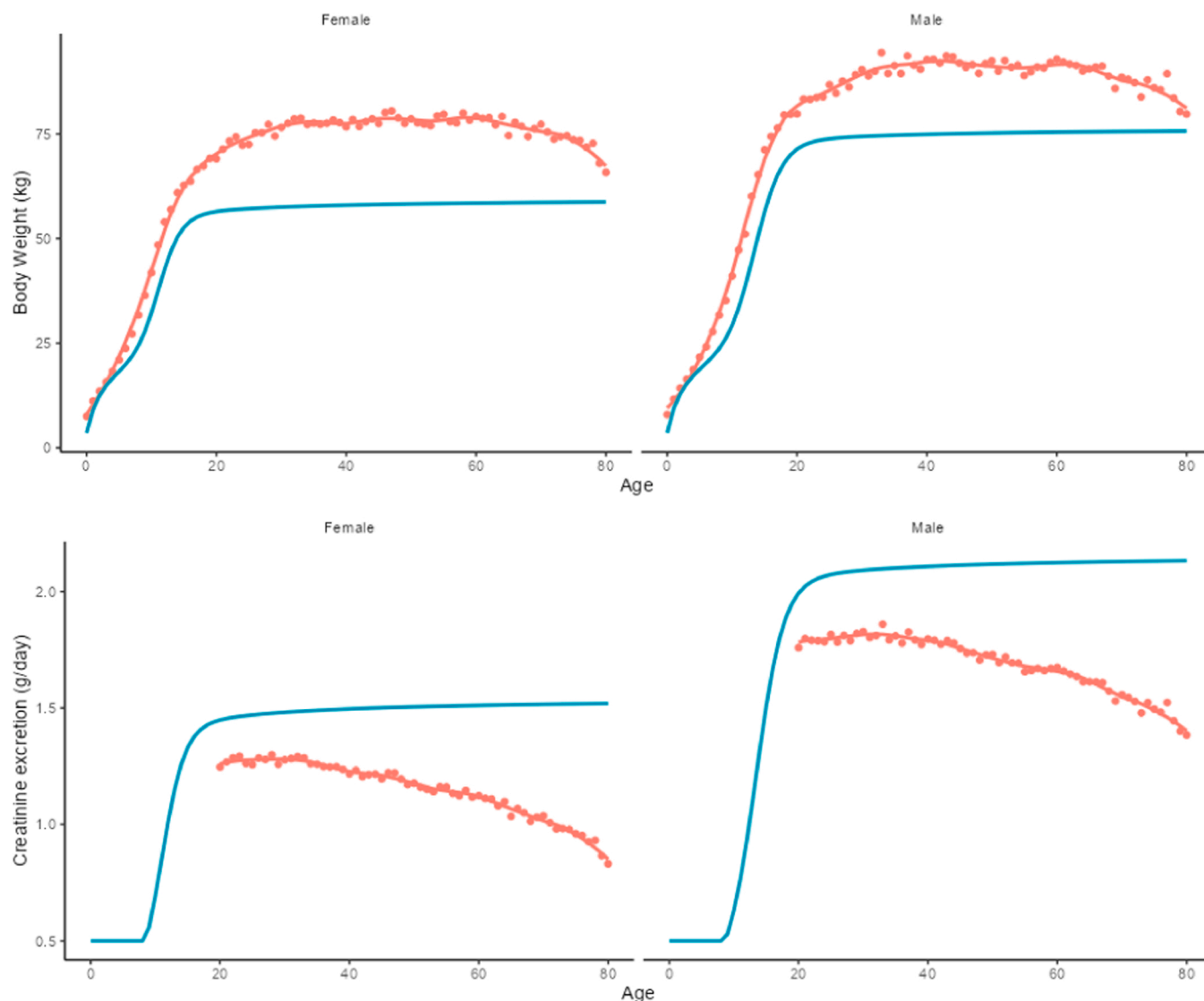


Fig. 4. Top: Body weight (kg) as a function of age (0–79 year) and gender, as modeled by O’Flaherty (1993) (green line), or estimated mean per year (red points) and loess smoothing (red line) as recorded in NHANES surveys 2003–2004–2017–2018. Bottom: Creatinine excretion (g/day) as a function of gender and age, as modeled by Choudhury et al. (2001) (green line), or as modeled by Ix et al. (2011)’s equation using NHANES 2003–2004 to 2017–2018 data (estimated mean at 20–79 year old: red points, corresponding smooth regression: red line). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

common covariates such as gender, age, and body weight (Barr et al., 2005), making the conversion at the population level of Cd excretion in $\mu\text{g/day}$ to $\mu\text{g/g}$ creatine particularly difficult. The creatinine excretion values obtained using the formula used in Choudhury et al. (2001) are much higher than the ones obtained using the Ix et al. (2011) derivation, which result in an underestimation of the U-Cd levels when converting the KN model output ($\mu\text{g/day}$) to U-Cd levels expressed as $\mu\text{g/g}$ creatinine and contribute to the underestimation observed with Model #1 in Fig. 2. Creatinine excretion has an even higher impact on U-Cd estimations expressed in $\mu\text{g/g}$ creatine for older adults, as creatinine excretion is estimated to decrease sharply with age and explains most of the differences observed between Model #1 and Model #2 in Fig. 2. The increase in Cd per g creatinine observed for individuals > 50 years old with Model #2 results from this decrease in creatinine excretion estimated with the Ix et al. (2011) model (Fig. 4) while the plateau reached with Model #1 (Fig. 3) is a result of the constant creatinine excretion estimated in the model used by Choudhury et al. (2001).

As demonstrated above, creatinine excretion data have a substantial impact on the results from the PBPK model. Yet, the inclusion of

creatinine in the model is simply an artifact of the biomonitoring process rather than a fundamental pharmacokinetic parameter. If U-Cd measurements could include a full day collection of urine rather than a spot sample, this “correction” parameter could be avoided, and the uncertainty associated with it reduced. NHANES only collects random spot urine samples. In a study by Akerstrom et al. (2014), Cd in 24-hr samples and first morning spot samples adjusted for dilution resulted in a high accuracy (<20% bias) in the observed slope coefficient. Biomarkers in spot urine samples showed to reach these accuracy levels with at least two repeated samples per participant, one sample per participant resulting in 40–50% accuracy. Because we use the estimated mean U-Cd for our subpopulation which includes a high number of individuals, these mean values are more robust estimates than the single spot samples.

Of note, all the PBPK models presented and available for Cd are population models, and no variability is considered (Kjellstrom and Nordberg, 1978). Although Béchaux et al. (2014) used the parameter values fitted to empirical data by Kjellstrom and Nordberg (1978) and their initially assumed ranges as parameters of triangular distributions

Table 3

Estimated dietary intake, assuming a constant intake per kg BW from birth to 50 years of age, leading to a certain mean urinary cadmium (U-Cd) excretion or mean concentration of cadmium in the kidney cortex (K-Cd) in a population of 50-year-old men and women, according to model #1, #2 or #3 (reverse dosimetry, see text for details).

Biomarker or tissue concentration	Women			Men		
	Model #1	Model #2	Model #3	Model #1	Model #2	Model #3
Mean U-Cd	0.47 µg/kg	0.47 µg/kg	0.56 µg/kg BW/day [CI 90%: 0.42;	1.07 µg/kg	0.88 µg/kg	0.98 µg/kg BW/day [CI 90%: 0.63;
1 µg/g creatinine*	BW/day	BW/day	0.76] µg/kg BW/day	BW/day	BW/day	1.49] µg/kg BW/day
Mean K-Cd	0.71 µg/kg	0.78 µg/kg	0.36 µg/kg BW/day [CI 90%: 0.28;	1.41 µg/kg	1.23 µg/kg	0.49 µg/kg BW/day [CI 90%: 0.33;
50 µg/g cortex	BW/day	BW/day	0.48] µg/kg BW/day	BW/day	BW/day	0.73] µg/kg BW/day

*Women: 1 µg/g creatinine corresponding to a U-Cd concentration of 1.18 µg/day assuming a creatinine excretion of 1.18 g creatinine/day as estimated by Equation D of [Ix et al. \(2011\)](#) in 50 year-old White women with a bodyweight of 79 kg; Men: 1 µg/g creatinine corresponding to a U-Cd concentration of 1.71 µg/day assuming a creatinine excretion of 1.71 g creatinine/day as estimated by Equation D of [Ix et al. \(2011\)](#) in 50 year-old in White men with a bodyweight of 91 kg

to model interindividual variability, the assumed ranges were not meant to reflect individual variability but uncertainty in the original model ([Kjellstrom and Nordberg, 1978](#)). The [Fransson et al. \(2014\)](#) study provides some measure of uncertainty around the most sensitive estimates but, because only one observation per compartment and individual was available in their study, the population variability was not estimated. As such, and until we obtain data on individual variability for the parameters of the KN model, these population estimates might need further adjustment factors to establish a toxicological reference value for the most vulnerable individuals. Having a credible interval around the mean provides more confidence in the estimated value of the mean, while still not representing variability.

3.2. Implementing the PBPK models in Cd forward and reverse dosimetry analyses

To understand the health implications of biological monitoring data in setting toxicological reference values, a forward dosimetry approach can be used where internal exposures are estimated from measurements of external exposure (toxicity studies). A reverse dosimetry approach (dose reconstruction) can also be used where estimates of external exposures are made based on biological monitoring data ([Ruiz and Fowler, 2015](#)). The forward and reverse dosimetry approaches make use of a PBPK model ([Ruiz and Fowler, 2015](#)). [Table 2](#) provides results of a forward dosimetry approach for each of the evaluated models (#1 through #3 describe in [Table 1](#)) assuming a constant mean ingested dose of 0.2 µg/kg BW/day from birth to 50 years of age. [Table 3](#) provides the results of a reverse dosimetry approach assuming a point of departure (POD) of 1 µg/g creatinine U-Cd. The reverse dosimetry can also be done in the application using a kidney cortex POD. Because the output of the KN model is in µg Cd excreted per day, in the reverse dosimetry analysis we need to convert the POD from µg/g creatinine to a µg/day by estimating the creatinine excretion (g/day) in the population on which the POD was obtained.

4. Conclusions

When determining the reference dose for chemical contaminants from a point of departure estimated from tissue concentrations or biomarkers, reverse dosimetry approaches make use of PBPK models. We reviewed and evaluated previously published Cd PBPK models for use in evaluating potential Cd toxicological reference values for the U.S. population. Of the three potential models evaluated for the U.S. population (models #1, 2 and 3), model #1 is not adapted for bodyweight, kidney weight or creatinine excretion and model #2 greatly underestimates the U-Cd: K-Cd ratio, while model #3 is adapted for the U.S. population in bodyweight, kidney weight and creatinine excretion, does not underestimate the U-Cd: K-Cd ratio and provides estimates with credible intervals around the mean to further inform regulatory decision making. We developed a web-based tool implementing the evaluated models and other potential adaptations to facilitate outcome comparisons of

forward and reverse dosimetry analyses using different models, assumptions, or inputs. Validated PBPK models which are fit for use on a specific subpopulation and where uncertainty estimates around the mean are included provide a higher degree of confidence when establishing reference dose concentrations for that specific subpopulation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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The authors declare they have nothing to disclose.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.toxlet.2022.07.812](https://doi.org/10.1016/j.toxlet.2022.07.812).

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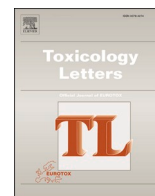
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Update

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Corrigendum

Corrigendum to “Cadmium physiologically based pharmacokinetic (PBPK) models for forward and reverse dosimetry: Review, evaluation, and adaptation to the U.S. population” [Toxicol. Lett. 367 (2022) 67–75]

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The authors regret label errors in Figure 1 and Tables S2 and S3. In Figure 1, the legends of the arrows starting from the “Daily uptake” compartment to Blood 1 and Blood 3 compartments were reversed: “1 - C7: 75% (or 1 - C8/I if $C7 \times I > C8$)” should be the legend of the arrow pointing to the Blood 1 compartment (and not to the “Blood 3” compartment), while “C7: 25% (with $C7 \times I < C8$: 1 μg)” should be the legend of the arrow pointing to the “Blood 3” compartment (and not to the “Blood 1” compartment). Additionally, the unit of C8 noted on this arrow should be “ μg ” and not “ $\mu\text{g}/\text{d}$ ”.

In supplementary Tables S2 and S3, C7 should be described as “Fraction of uptake pool transferred to blood pool B1” instead of “Fraction of uptake pool transferred to blood pool B3”. Additionally, in

tables S2 and S3, C13 is the rate of transfer in liver to blood pool B1 (B1 was not specified in the table S2), C16 is the rate of transfer of Cd in blood pool B2 to blood pool B3 (B3 was not specified in the tables S2 and S3) and C18 is the rate of transfer of Cd in kidney to blood pool B1 (B1 not specified in the table S2).

The authors would like to underscore that these corrections only affect the description of the model in Figure 1 and Tables S2 and S3. The model, code, results, and tool are not affected and thus, as such, these changes have no consequence on the results, discussion, and conclusions of the manuscript.

The authors would like to apologize for the inconvenience caused by these errors. A corrected version of Figure 1 is provided below.

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